WEI XUAN

xuanwei@suat-sz.edu.cn

Research Interests

My research focuses on advancing **AI computility** through high-performance, energy-efficient, and trustworthy AI computing systems. I explore novel **AI accelerator architecture**, **compiler optimization**, and **hardware security**, emphasizing **cross-layer co-design and optimization** to enhance computing performance, energy efficiency, and trustworthiness.

Working Experience

Shenzhen University of Advanced Technology 2026.06 – Present

- Tenure-Track Assistant Professor in Computility Microelectronics

The Hong Kong University of Science and Technology 2023.06 – 2026.06

- Post-doctoral Fellow in AI Chip Center for Emerging Smart Systems
- Advisors: Prof. Zhongrui Wang and Prof. Tsung-Yi Ho

Education

Institute of Computing Technology, Chinese Academy of Sciences 2016.09 – 2023.01

- Ph.D. in Computer System Architecture
- Advisors: Prof. Zhimin Tang, Prof. Dongrui Fan

Northeastern University 2012.09 – 2016.06

- B.E. in Computer Science and Technology

Publications

TCAD'26 CCF-A
Wei Xuan, Zihao Xuan, Rongliang Fu, Ning Lin, Kwunhang Wong, Zikang Yuan, Lang Feng, Zhongrui Wang, Tsung-Yi Ho, Yuzhong Jiao, Luhong Liang, "Towards Secure and Efficient DNN Accelerators via Hardware-Software Co-Design", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026.

TCAD'26 CCF-A
Rongliang Fu, Ran Zhang, Libo Shen, **Wei Xuan**, Ning Lin, Junying Huang, Bei Yu, Tsung-Yi Ho, "Efficient Multi-Array Parallel Scheduling for In-Memory SIMD Computations", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026.

ICCAD'26 CCF-B
Kwunhang Wong, Jichang Yang, Karl M.H. Lai, Hegan Chen, Songqi Wang, **Wei Xuan**, Ning Lin, Han Wang, Xiaojuan Qi, Zhongrui Wang, "RRAM-DP: Device-Calibrated Differential Privacy for In-Memory Edge Learning", *In Proceedings of the 45th IEEE/ACM International Conference On Computer Aided Design*, 2026.

DATE'26 CCF-B
Rongliang Fu, **Wei Xuan**, Shuo Yin, Guangyu Hu, Chen Chen, Hongce Zhang, Bei Yu and Tsung-Yi Ho, "eLogic: A E-Graph-based Logic Rewriting Framework for Majority-Inverter Graphs", *In Proceedings of Design, Automation & Test in Europe Conference*, 2026.

DAC'25 CCF-A
Wei Xuan, Zhongrui Wang, Lang Feng, Ning Lin, Zihao Xuan, Rongliang Fu, Tsung-Yi Ho, Yuzhong Jiao, Luhong Liang, "SeDA: Secure and Efficient DNN Accelerators with Hardware/Software Synergy", *In Proceedings of the 62nd Annual ACM/IEEE Design Automation Conference*, 2025.

DAC'25 CCF-A
Zihao Xuan, Yuxuan Yang, **Wei Xuan**, Zijia Su, Song Chen, Yi Kang, "YOCO: A Hybrid In-Memory Computing Architecture with 8-bit Sub-PetaOps/W In-Situ Multiply Arithmetic for Large-Scale AI", *In Proceedings of the 62nd Annual ACM/IEEE Design Automation Conference*, 2025.

ICCAD'24 CCF-B
Ning Lin, Songqi Wang, Xinyuan Zhang, Shaocong Wang, Yangu He, Woyu Zhang, Bo Wang, Jiankun Li, Mingzi Li, Binbin Cui, Yi Li, Jia Chen, Chunwei Xia, **Wei Xuan**, Xiaoming Chen, Dashan Shang, Zhongrui Wang, "LSMR: Synergy Randomness in Liquid State Machine and RRAM-based Analog-digital Accelerator", *In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design*, 2024.

SRDS'24 CCF-B
Wei Xuan, Yan Liang, Huawei Cao, Ning Lin, Xiaochun Ye, Dongrui Fan, "DTC: Real-Time and Accurate Distributed Triangle Counting in Fully Dynamic Graph Streams", *In Proceedings of the 43rd International Symposium on Reliable Distributed Systems*, 2024.

IJSEKE'21 CCF-C
Wei Xuan, Huawei Cao, Mingyu Yan, Zhimin Tang, Xiaochun Ye, Dongrui Fan, "BSR-TC: Adaptively Sampling for Accurate Triangle Counting over Evolving Graph Streams", *International Journal of Software Engineering and Knowledge Engineering*, 2021.

SEKE'21 CCF-C
Wei Xuan, Huawei Cao, Mingyu Yan, Zhimin Tang, Xiaochun Ye, Dongrui Fan, "Triangle Counting by Adaptively Resampling over Evolving Graph Streams", *In Proceedings of the 33rd International Conference on Software Engineering & Knowledge Engineering*, 2021.

ISEC'19
Wei Xuan, Guangming Tang, Peiyao Qu, Zhimin Tang, Xiaochun Ye, Dongrui Fan, Zhimin Zhang, Ninghui Sun, "Logic Design of a 16-bit Bit-Slice Shifter for 64-bit RSFQ Microprocessors", *In Proceedings of the 17th IEEE International Superconductive Electronics Conference*, 2019.

PATENTS

Application granted CN109308190B
Wei Xuan, Xiaochun Ye, Dongrui Fan, Hao Zhang, Da Wang, "Shared Row Buffer System and Shared Row Buffer based on 3D Stacked Memory Architecture".

Application granted CN111049503B
Zhiming Zhang, Guangming Tang, **Wei Xuan**, Kuozhong Zhang, Peiyao Qu, Jiahong Yang, "A Superconducting Flip-flop and Its Operation Method".

Application granted CN112015472B
Xinxin Wu, Zhihua Fan, **Wei Xuan**, Wenmin Li, Xiaochun Ye, Dongrui Fan, "A Method and System for Accelerating Sparse Convolutional Neural Networks Based on a Dataflow Architecture".

Application granted CN112015473B
Xinxin Wu, Zhihua Fan, **Wei Xuan**, Wenmin Li, Xiaochun Ye, Dongrui Fan, "A Method and System for Accelerating Sparse Convolutional Neural Networks Based on a Dataflow Architecture".

Application granted CN111147045B
Zhiming Zhang, Guangming Tang, Kuozhong Zhang, **Wei Xuan**, Peiyao Qu, Jiahong Yang, "A Method and System for Resetting a Superconducting Circuit".

HONORS & AWARDS

Best Paper Runner Up SRDS 2024

Best Paper Award: Second Place SEKE 2021

National Scholarship University of Chinese Academy of Sciences, 2021

Outstanding Student University of Chinese Academy of Sciences, 2019

First Prize Academic Scholarship University of Chinese Academy of Sciences, 2017

Best System Control Award International Aerial Robotics Competition, 2014

Best Mission Planning Award International Aerial Robotics Competition, 2014

National Encouragement Scholarship Northeastern University, 2013

First Prize Scholarship Northeastern University, 2013

Research Talks

Invited Seminar 2026.05	Southern University of Science and Technology & Ant Group Joint Research Platform for Trusted Intelligent Computing, "Software-Hardware Co-Optimization for Secure AI Accelerated Computing".
Invited Speaker 2026.01	Yuelu Forum, Hunan University, "Trusted and Efficient AI Accelerated Computing Enabled by Software-Hardware Co-Optimization".
Invited Speaker 2025.12	16th International Young Scholars Forum, Sun Yat-sen University, "Software-Hardware Co-Optimization for Trusted and Efficient AI Accelerated Computing".
Invited Speaker 2025.12	13th Shenzhou Forum for Outstanding Young Scholars, Harbin Institute of Technology, "Software-Hardware Co-Optimization for Trusted and Efficient AI Accelerated Computing".
Invited Speaker 2025.10	Integrated Circuit Doctoral Forum, Shandong University, "Domain-Specific Acceleration Targeting Efficient AI".

Academic Service

Technical Program Committee	44th IEEE International Conference on Computer Design (ICCD 2026) 32nd Asia and South Pacific Design Automation Conference (ASP-DAC 2027)
Session Chair (Appointed)	44th IEEE International Conference on Computer Design (ICCD 2026)
Artifact Evaluation Committee	59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)
Reviewer	IEEE Transactions on Information Forensics and Security (TIFS) ACM Transactions on Knowledge Discovery from Data (TKDD)